2026/4/13

习题11.1 2, 3,6,7,
习题11.2 10,11,12每道大题中任选3道小题, 其余所有大题中任选10道。
习题11.3 1，其余所有大题中任选5道。
习题11.4 任选3道大题
习题11.5 2,3,4,5中共任选6道小题 其余所有大题中共任选3道。

11.1

11.1.2

Discuss the convergence of

$$\sum_{n=1}^{\infty} \frac{1}{(n+1)^{\alpha} n^{\beta}} - \frac{1}{n^{\alpha}(n+1)^{\beta}}, \alpha > 0, \beta > 0.$$

Solution: WLOG assume $\alpha < \beta$ ($\alpha = \beta$ clearly converges). Then

$$\begin{aligned} \frac{1}{(n+1)^{\alpha} n^{\beta}} - \frac{1}{n^{\alpha}(n+1)^{\beta}} &= \frac{1}{(n(n+1))^{\alpha}} \left(\frac{1}{n^{\beta-\alpha}} - \frac{1}{(n+1)^{\beta-\alpha}} \right) \\ &\sim n^{-2\alpha} \cdot \frac{\beta-\alpha}{n^{\beta-\alpha+1}} = (\beta-\alpha) n^{-1-\alpha-\beta}. \end{aligned}$$

Hence it converges (absolutely) for all $\alpha, \beta > 0$.

11.1.3

Suppose $\{a_n\}, \{b_n\}$ are monotone and bounded, discuss the convergence of $\sum_{n=2}^{\infty} (a_n b_{n-1} - b_n a_{n-1})$.
Solution: Suppose $c_n = a_n - a_{n-1}$ and $d_n = b_n - b_{n-1}$, then $\sum c_n$ and $\sum d_n$ converges, and $a_n b_{n-1} - b_n a_{n-1}$ $= c_n b_{n-1} - a_{n-1} d_n$. Note that $\sum_{n=1}^{\infty} |c_n b_{n-1}| \leqslant M |\sum_{n=1}^{\infty} c_n|$ where b_n is bounded by M, hence the series converges. Likewise $\sum |a_{n-1} d_n|$ converges, so $\sum_{n=2}^{\infty} a_n b_{n-1} - b_n a_{n-1}$ converges absoutely.

11.1.6

Suppose $\sum a_n^2$ converges, prove that $\sum a_n^3$ converges absolutely.
Proof: WLOG suppose $a_n \geqslant 0$. Since $\sum a_n^2$ converges, $a_n \to 0$ and take N such that $\forall n > N, |a_n| < 1$. Then $\sum_{n>N} a_n^3$ $\leqslant \sum_{n>N} a_n^2 < \infty$ so the series converges.

11.1.7

Give examples of:
(1) $\sum_{n=1}^{\infty} a_n$ converges but not $\sum a_n^3$;
(2) $\sum a_n^3$ converges but not $\sum \frac{a_n}{n}$.
Solution: (1) Let $a_{3n-2} = \frac{1}{n^{1/3}}$, $a_{3n-1} = a_{3n} = -\frac{1}{2n^{1/3}}$. Then $a_{3n+1} + a_{3n+2} + a_{3n+3} = 0$ and $a_n \to 0$. Hence $\sum a_n$ converges. However, $a_{3n-2}^3 + a_{3n-1}^3 + a_{3n}^3 = \frac{3}{4n}$, so $\sum a_n^3$ diverges.
(2) Let $a_n = b_n / \log n$ where $b_{3n} = -1, b_{3n-1} = b_{3n+1} = 2^{-1/3}$, then by Dirichlet test $\sum a_n^3$ converges. However $\frac{a_{3n}}{3n}$ $+ \frac{a_{3n+1}}{3n+1} + \frac{a_{3n+2}}{3n+2} > \frac{C}{n \log n}$, so $\sum a_n / n$ diverges.

11.2

11.2.1

Give examples: when $\limsup a_n^{1/n} = 1$, $\sum a_n$ can both converge and diverge.
Solution: $a_n = \frac{1}{n}$ => $\sum a_n$ diverges; $a_n = \frac{1}{n \log^2 n}$ => $\sum a_n$ converges.

11.2.2

Suppose $a_1 \geqslant a_2 \geqslant \cdots \geqslant 0$. Prove that if $\sum a_n$ converges, then $\lim_{n\to\infty} na_n = 0$.
Proof: For any $\varepsilon > 0$, take N such that $\sum_{m>N} a_m < \varepsilon$. Then for $n > N$, $\varepsilon > \sum_{N<m\leqslant n} a_m \geqslant (n-N)a_n$. Hence $\limsup na_n < \varepsilon$ for any $\varepsilon > 0$. Therefore $\lim_{n\to\infty} na_n = 0$.

11.2.3

(1) Suppose $\{a_n\}$ decreases and tends to 0, $\sum a_n$ diverges. Prove that $\sum \min\left\{a_n, \frac{1}{n}\right\}$ diverges.
(2) If both $\{a_n\}$, $\{b_n\}$ decreases and tends to 0, $\sum a_n, \sum b_n$ diverges. It is necessary that $\sum \min\{a_n, b_n\}$ diverges?
Proof: (1) By Ex11.2.2 if $\sum \min\left\{a_n, \frac{1}{n}\right\}$ converges, then $\min\left\{a_n, \frac{1}{n}\right\} = o\left(\frac{1}{n}\right)$, so $\min\left\{a_n, \frac{1}{n}\right\} = a_n$ for some $n > N$. Hence $\sum a_n$ converges, a contradiction.
(2) No. Consider $N_k = 3^{2^k}$, $c_k = 1/N_k^2$. For $N_k \leqslant n < N_{k+1}$, if k is odd, let $a_n = c_{2k-1}$, $b_n = c_{2k}$; if k is even, let $a_n = c_{2k}$, $b_n = c_{2k-1}$.
Then $\sum a_n = \sum b_n$ diverges, but $\sum \min\{a_n, b_n\}$ converges, since the sum over each block is 1 for c_{2k-1} and $3^{-2^{k-1}}$ for c_{2k}.

11.2.4

Is there a convergent $\{a_n\}$ such that $a_n \geqslant 0$ and $a_n \geqslant \frac{1}{n}$ infinitely?
Solution: Yes, e.g. $a_{2^n} = \frac{1}{2^n}$ and otherwise $a_n = 0$.

11.2.5

Suppose $\sum a_n$ converges for $a_n \geqslant 0$, $a = \lim_{n\to\infty} na_n$. Prove that $a = 0$.
Proof: If $a > 0$, then there exists N such that $na_n > a/2$ for all $n > N$. Hence $\sum a_n > \sum \frac{a}{2n} = \infty$, a contradiction.

11.2.6

Determine whether $\sum \left(\frac{(2n-1)!!}{(2n)!!}\right)^p$ converges.
Solution: Recall that $a_n = \frac{(2n-1)!!}{(2n)!!} \sim \frac{1}{\sqrt{\pi n}}$, so it converges when $p > 2$.

11.2.7

For $\alpha > 0$, determine the order of $\frac{n!}{(n+\alpha)\cdots(1+\alpha)}$.
Solution: Recall that

$$\Gamma(s) = \lim_{n\to\infty} \frac{n! n^s}{(n+s)\cdots s}.$$

Hence $\frac{n!}{(n+\alpha)\cdots(1+\alpha)} = \alpha\Gamma(\alpha) \cdot n^{-\alpha}$.

11.2.8

Determine convergence of $\sum \frac{1}{n^p (\log n)^q (\log\log n)^r}$.
Solution: $p > 1$ => converge, $p < 1$ => diverge.
If $p = 1$, by substituting $n \to 2^n$, $q > 1$ => converge, $q < 1$ => diverge.
If $p = q = 1$, likewise $r > 1$ => converge, $r \leqslant 1$ => diverge.

11.2.9

Suppose $\{a_n\}$ decreasing positive sequence, $\sum a_n$ converges, prove that $\sum a_n e^{-a_n/a_{n+1}}$ converges.
Proof: Trivial since $e^{-a_n/a_{n+1}} \leqslant e^{-1}$.

11.2.10

Determine convergence:
(1) $\sum \frac{1}{n^{1+1/\log\log n}}$.
Solution: $n^{1/\log\log n} = \exp(\log n/\log\log n) \geqslant \exp(2\log\log n) = \log^2 n$ for large n, so the series <= $\sum \frac{1}{n\log^2 n}$ which converges.
(2) $\sum \frac{1}{n^{1+(\log\log n)/\log n}}$.
Solution: $n^{(\log\log n)/\log n} = \exp\log\log n = \log n$, so $\sum \frac{1}{n\log n}$ diverges.
(3) $\sum \frac{1}{n^{1+2(\log\log n)/\log n}}$.
Solution: Likewise = $\sum \frac{1}{n\log^2 n}$ converges.
(4) $\sum \frac{1}{(\log n)^{\log n}}$.
Solution: For large n, $(\log n)^{\log n} > (e^2)^{\log n} = n^2$, so it converges.
(5) $\sum \frac{1}{(\log n)^{\log\log n}}$.
Solution: Note that $(\log\log n)^2 < \log n$ for large n, so $(\log n)^{\log\log n} < n$. Hence it diverges.
(6) $\sum \frac{1}{(\log\log n)^{\log n}}$.
Solution: For large n, $\log n \cdot \log\log\log n > 2\log n$, so $(\log\log n)^{\log n} > n^2$ hence it converges.
(7) $\sum \frac{1}{a^{\log n}}$. $a > 1$.
Solution: $a^{\log n} = n^{\log a}$ converges iff $a > e$.
(8) $\sum \frac{1}{n^{\sqrt[n]{n}}}$.
Solution: verify that $\lim_{n\to\infty} n^{\sqrt[n]{n}-1} = 1$, and $\sum \frac{1}{n}$ diverges, so the series also diveges.

11.2.11

(1) $\sum \frac{1}{\sqrt{n+n^2}}$ diverges.
(2) $\sum 2^n \sin\frac{\pi}{3n}$ diverges.
(3) $\sum \left(1-\cos\frac{1}{n}\right)$ converges.
(4) $\sum \frac{1}{n}\tan\frac{1}{n}$ converges.
(5) $\sum \frac{1}{\sqrt{n^3+1}}$ converges.
(6) $\sum \frac{\log n}{n^p}$ converges if $p > 1$.

11.2.12

(1) $\sum \frac{(2n-1)!!}{n!}$ diverges.
(2) $\sum \frac{n!}{n^n}$ converges.
(3) $\sum \frac{n^2}{2^n}$ converges.
(4) $\sum (\sqrt[n]{a-1})^p$ diverges.
(5) $\sum \left(1-n\sin\frac{1}{n}\right)^p$ converges if $p > 1/2$.
(6) $\sum \left(\frac{n}{2n+1}\right)^n$ converges

11.2.13

Suppose $a_n > 0$, $\sum a_n$ converges. Prove that
(i) $\sum a_n^{1-1/(kn+1)}$ converges for $k > 0$.
(ii) If $b_n \log n \to 0$, then $\sum a_n^{1-b_n}$ converges.
Proof (ii) WLOG assume $a_n < 1$. Let $A = \left\{n : a_n \leqslant \frac{1}{n^2}\right\}$ and $B = \{n : a_n > 1/n^2\}$. Then $\sum_{n\in A} a_n^{1-b_n}$ converges since $1 - b_n > \frac{3}{4}$ for large n, so $a_n^{1-b_n} \geqslant \frac{1}{n^{3/2}}$ and the sum converges.
For $n \in B$, $a_n^{1-b_n} = \exp((1-b_n)\log a_n)$, where $0 > \log a_n > -2\log n$, so $b_n(-\log a_n) < 2b_n \log n \leqslant e^2$. Hence $a_n^{1-b_n} \leqslant e^2 a_n$, so $\sum_{n\in B} a_n^{1-b_n}$ also converges.

11.2.14

Prove that $\sum \frac{1}{n^{1+1/\log n}}$ diverges, $\sum \frac{1}{n^{1+1/\sqrt{\log n}}}$ converges, $\sum \frac{1}{n^{1+|\sin n|}}$ diverges.
Proof: (1) $n^{1/\log n} = e$ so $\sum \frac{1}{n^{1+1/\log n}}$ diverges.
(2) $n^{1/\sqrt{\log n}} = e^{\sqrt{\log n}} < e^{\frac{1}{2}\log n}$ so $\sum \frac{1}{n^{1+1/\sqrt{\log n}}} < \sum \frac{1}{n\sqrt{n}}$ converges.
(3) Take the sequence of continued fractions of $\theta = \frac{1}{\pi}$: $p_k/q_k \to \theta$. Then $\left|\theta - \frac{p_k}{q_k}\right| \leqslant \frac{1}{q_k q_{k+1}}$.
Note that $n^{|\sin n|} = \exp(\log n \cdot |\sin n|) \leqslant \exp(\pi \log n \cdot \|n\theta\|)$. We consider $\sum_{n=N}^{2N} n^{-1-|\sin n|}$ $\geqslant \sum_{n=N}^{2N} \frac{1}{2N} e^{-\pi \log 2N \cdot \|n\theta\|}$.
Case1: For large N, there always exist $q = q_k$ such that $C \log N \leqslant q_k \leqslant N$. Then we partition $(N, 2N]$ into $M = \lfloor N/q \rfloor$ blocks of length q. In each block, the sum $\geqslant \frac{1}{2N} \sum_{j=1}^{q} e^{-\pi \log 2N \cdot (j-1)/q}$, which is greater than $\frac{1}{2N} \cdot \frac{1}{1-e^{-\pi \log(2N)/q}}$. Since q $\geqslant C \log N$, this part is bounded by $\frac{q}{2\pi N \log N}$, so summing up we obtain the partial sum $\geqslant \frac{1}{2\pi \log N}$. Let $N = 2^m$ and sum over m, we obtain $\sum \frac{1}{n^{1+|\sin n|}} \geqslant \sum \frac{c}{m} = \infty$.
Case2: Infinitely many $q_{k+1} \geqslant e^{q_k/C}$. Consider $n = jq_k$ for $j \leqslant \frac{1}{\|q_k\theta\| \log(1/\|q_k\theta)}$, then $\|n\theta\|$ is bounded by j/q_{k+1}, so $\sum \frac{1}{n^{1+|\sin n|}} \geqslant \frac{\log q_{k+1}}{q_k} \geqslant \frac{1}{C}$. Hence summing up these n in (q_k, q_{k+1}), the series diverges.

11.2.15

Prove that $\sum_p \frac{1}{p}$ diverges.
Proof: Euler gives $\zeta(s) = \sum n^{-s} = \prod (1 - p^{-s})$. Taking log we obtain $\log \zeta(s) = \sum p^{-s} + O(1)$. Since ζ has a simple pole at $s = 1$, $\sum p^{-1}$ diverges.

11.2.16 & 17

Given a string of numerals α, $A = \{n \in \mathbb{N} : \text{the decimal } n \text{ does not contain } \alpha\}$. Prove that $\sum_{n \in A} n^{-1}$ converges.
Proof: Suppose α has length k, then for decimals of length nk, there are at most $(10^k - 1)^n$ possible numbers in A.
Hence $\sum_{n \in A} n^{-1} \leqslant \sum_n \frac{(10^k-1)^n}{10^{kn}}$ converges.

11.2.18

Suppose $a_n > 0$, if exists ℓ such that for large n, $(n - \ell)\left(\frac{a_n}{a_{n+1}} - 1\right) \leqslant 1$, then $\sum a_n$ diverges.
Proof: Just apply Rabbe's test to $\{a_{n-N}\}$ for $N > \ell$.

11.2.21

Suppose $a_n > 0$, $\sum a_n$ converges, prove that $\sum H_n$, $\sum A_n$ and $\sum G_n$ converge.
Proof: We have $\sum H_n \leqslant 4 \sum a_n$, and $\sum G_n \leqslant e \sum a_n$. $\sum A_n$ always diverge.

11.2.22

There cannot be $a_n > 0$ such that $\sum a_n$ and $\sum \frac{1}{n^2 a_n}$ converge.
Proof: Otherwise $\sum \frac{2}{n} \leqslant \sum a_n + \frac{1}{n^2 a_n}$ converges.

11.2.23

Suppose $\alpha > 0$, prove that exists $a_n > 0$ such that $\sum a_n$ and $\sum \frac{1}{n^{2+\alpha} a_n}$ converge.
Proof: Take $a_n = n^{-\beta}$, such that $1 < \beta < 1 + \alpha$.

11.2.24&25

Suppose $\alpha > 0$, prove that exists $a_n > 0$ such that $\sum a_n$ and $\sum \frac{1}{n^2 (\log^{2+\alpha} n) a_n}$ converge.
Proof: Let $a_n = \frac{1}{n \log^\beta n}$ where $1 < \beta < 1 + \alpha$.

11.2.26

(1) If $a_n \to 0$ decreasing, then $\lim_{n\to\infty} a_n = 0$ iff $\sum \left(1 - \frac{a_{n+1}}{a_n}\right)$ diverges.
(2) Suppose $a_n > 0$ increasing, then $\{a_n\}$ is bounded iff $\sum \left(1 - \frac{a_n}{a_{n+1}}\right)$ converges.
(3) $\sum a_n$ converges iff $\sum \frac{a_n}{S_n}$ converges.
Proof: (1) Note that $\sum_{n=N}^{N+K} 1 - \frac{a_{n+1}}{a_n} \leqslant \log a_N - \log a_{N+K}$ so if the series diverges then $\lim_{n\to\infty} a_n = 0$. If $\sum \left(1 - \frac{a_{n+1}}{a_n}\right)$ converges, then $x_n = 1 - \frac{a_{n+1}}{a_n} \sim -\log(1 - x_n)$. Hence $\sum -\log(1 - x_n) = \lim_{n\to\infty} \log a_0 - \log a_n$ diverges, so $a_n \to 0$.
(2) is the same as (1)
(3) If $\sum a_n$ converges, then the partial sum $\sum_{n=N}^{N+K} \frac{a_n}{S_n} \leqslant \frac{S_{N+K} - S_{N-1}}{S_N} \leqslant \frac{S}{S_N} - 1$, which tends to 0 as $N \to \infty$.
If $\sum a_n$ diverges, then $\sum_{n=N}^{N+K} \frac{a_n}{S_n} \geqslant \frac{S_{N+K} - S_{N-1}}{S_{N+K}} \to 1$ as $K \to \infty$, so $\sum \frac{a_n}{S_n}$ diverges.

11.2.28

Discuss the convergence of the hypergeometric series:

$$F(\alpha, \beta, \gamma; x) = 1 + \sum_{n=1}^{\infty} \frac{\alpha^{\underline{n}} \beta^{\underline{n}}}{n! \gamma^{\underline{n}}} x^n.$$

Solution:

- $|x| < 1$: absolutely converge
- $|x| > 1$: diverge
- $x = 1$:
  - $\gamma - \alpha - \beta > 0$: absolutely converge
  - $\gamma - \alpha - \beta \leqslant 0$: diverge
- $x = -1$:
  - $\gamma - \alpha - \beta > 0$: absolutely converge
  - $\gamma - \alpha - \beta \in (-1, 0]$: conditionally converge
  - $\gamma - \alpha - \beta \leqslant -1$: diverge

11.2.29

Suppose $\sum a_n$ diverges, $a_n > 0$. Determine the convergence of:
(1) $\sum \frac{a_n}{1+a_n}$; (2) $\sum \frac{a_n}{1+n^2 a_n}$; (3) $\sum \frac{a_n}{1+a_n^2}$; (4) $\sum \frac{a_n}{1+n a_n}$.
Solution: (1) Diverge. If infinitely many $a_n > 1$, then $\frac{a_n}{1+a_n} > \frac{1}{2}$ so it diverges. Otherwise for large n, $\frac{a_n}{1+a_n} > \frac{a_n}{2}$ so it diverges.
(2) Converge. $\frac{a_n}{1+n^2 a_n} < \frac{1}{n^2}$ which converges.
(3) Both are possible. $a_n = 1$ => diverge; $a_n = 2^n$ => converge.
(4) Both are possible. $a_n = 1$ => diverge; $a_{2^n} = 1$ otherwise $a_n = 0$ => converge.

11.3

11.3.1

Discuss whether the following series converge:
(1)

$$\sum \frac{(-1)^{\lfloor\sqrt{n}\rfloor}}{\sqrt{n}} = \sum_k (-1)^k \sum_{n=k^2}^{k^2+2k} \frac{1}{\sqrt{n}}.$$

Let $a_k = \sum_{n=k^2}^{k^2+2k} n^{-1/2}$ then $a_k \geqslant 2k/(k+1) > 1$ so the series diverges.
(2)

$$\sum \frac{(-1)^{\lfloor\sqrt{n}\rfloor}}{n} = \sum_k (-1)^k \sum_{n=k^2}^{k^2+2k} \frac{1}{n}.$$

Let $a_k = \sum_{n=k^2}^{k^2+2k} \frac{1}{n}$, then

$$|a_n - a_{n+1}| \leqslant \frac{2}{k^2} + \sum_{n=k^2}^{k^2+2k} \frac{1}{n} - \frac{1}{n+2k+1} \leqslant \frac{2}{k^2} + \frac{(2k+1)^2}{k^4} \leqslant \frac{10}{k^2}.$$

Hence $\sum (-1)^k a_k$ converges.
(3)

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n} + (-1)^n}.$$

Note that $\frac{1}{\sqrt{n}-1} - \frac{1}{\sqrt{n+1}+1} \geqslant \frac{1}{\sqrt{n}-1} - \frac{1}{\sqrt{n}+1} = \frac{2}{n-1}$, where $\sum \frac{2}{n-1} = \infty$. Hence the series diverges.
(4)

$$\sum \frac{(-1)^{\lfloor\sqrt{n}\rfloor}}{\sqrt{n+(-1)^n}} = \sum (-1)^k \sum_{n=k^2}^{k^2+2k} \frac{1}{\sqrt{k+(-1)^k}}.$$

Note that $\sum_{n=k^2}^{k^2+2k} \frac{1}{\sqrt{k+(-1)^k}} \geqslant \frac{2k}{k+1} > 1$, so the series diverges.
(5)

$$\sum \frac{(-1)^n}{\sqrt{n} + (-1)^{\lfloor\sqrt{n}\rfloor}}.$$

Let $c_n = (-1)^{\lfloor\sqrt{n}\rfloor}$, then $\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n}+c_n} = \frac{c_n}{\sqrt{n}(\sqrt{n}+c_n)}$. Clearly $\sum (-1)^n/\sqrt{n}$ converges, we show that $\sum \frac{(-1)^n c_n}{\sqrt{n}(\sqrt{n}+c_n)}$ converges. Likewise $\frac{1}{n} - \frac{1}{\sqrt{n}(\sqrt{n}+c_n)} = \frac{c_n}{n(\sqrt{n}+c_n)}$, and clearly $\sum \frac{(-1)^n c_n^2}{n(\sqrt{n}+c_n)} = \sum \frac{(-1)^n}{n(\sqrt{n}+c_n)}$ converges. It suffices to show $\sum \frac{(-1)^n c_n}{n}$ converges.
For $c_{2n} = c_{2n+1}$, $\frac{(-1)^{2n} c_{2n}}{2n} + \frac{(-1)^{2n+1} c_{2n+1}}{2n+1}$ is bounded by $\frac{1}{n^2}$, so this part converges. For $c_{2n} \neq c_{2n+1}$, i.e. $2n+1 = m^2$, $\frac{(-1)^{2n} c_{2n}}{2n+1} + \frac{(-1)^{2n+1} c_{2n+1}}{2n+1}$ is bounded by $\frac{4}{m^2}$, so this part also converges. Therefore the sum $\sum \frac{(-1)^n c_n}{n}$ converges.
(6)

$$\sum \frac{(-1)^{\lfloor\sqrt{n}\rfloor}}{\sqrt{n+(-1)^{\lfloor\sqrt{n}\rfloor}}} = \sum_k (-1)^k \sum_{n=k^2}^{k^2+2k} \frac{1}{\sqrt{n+(-1)^k}}.$$

Likewise the partial sum $\sum_{n=k^2}^{k^2+2k} \frac{1}{\sqrt{n+(-1)^k}}$ does not tend to 0, so the series diverges.

11.3.2

If for any $\{x_n\} \to 0$, $\sum a_n x_n$ converges, prove that $\sum |a_n|$ converges.
Proof: WLOG assume $a_n \geqslant 0$. If $\sum a_n$ diverges, then inductively define k_n as the least $k_n \geqslant 0$ such that $\sum_{i=k_{n-1}+1}^{k_n} a_i > n$. Let $x_i = 1/n$ for $k_{n-1} < i \leqslant k_n$. Then $\sum a_n x_n$ diverges while $x_i \to 0$, a contradiction. Hence $\sum |a_n|$ converges.

11.3.3

Prove that: if $\sum a_n^2 = \infty$, then exists $\{b_n\}$ such that $\sum b_n^2 < \infty$ but $\sum a_n b_n = \infty$.
Proof: WLOG assume $a_n \geqslant 0$. If a_n is unbounded, the result is trivial, just take a subsequence $a_{n_k} \to \infty$ and define $b_{n_k} = 1/ka_{n_k}$ and $b_n = 0$ otherwise.
Now suppose a_n is bounded by M. Inductively define N_i such that $\sum_{n=N_{i-1}+1}^{N_i} a_n \geqslant 1$, taking the least N_i, and for $N_{i-1} < n \leqslant N_i$, define $b_n = a_n/i$. Then $\sum a_n b_n \geqslant \sum 1/i = \infty$, while $\sum b_n^2 \leqslant (M+1)\sum 1/i^2 < \infty$.

11.3.4

Suppose $\lim_{n\to\infty} a_n = 0$ and a_n is monotone. Prove that:
(1) If $\sum a_n$ converges, then $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converges absolutely for any x.
(2) If $\sum a_n$ diverges, then $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converges conditionally for any x.
Proof: (1) trivial since $|\sin nx| \leqslant 1$.
(2) this is clearly incorrect, e.g. take $x = 0$ then $\sum a_n \cos nx = \sum a_n = \infty$.

11.3.5

Suppose $\sum a_n$ converges conditionally, and $S_n = S_n^+ + S_n^-$. Prove that $\lim_{n\to\infty} S_n^+/S_n^-$ exists and determine its value.
Proof: Suppose $A = \sum a_n = \lim_{n\to\infty} S_n$, then $S_n^+ + S_n^- \to A$ and $T_n = S_n^+ - S_n^- \to \infty$. Hence $\lim_{n\to\infty} \frac{S_n^+}{S_n^-} = \lim_{n\to\infty} \frac{S_n+T_n}{S_n-T_n} = -1$.

11.3.6

Suppose $b_n > 0$, $\lim_{n\to\infty} n(\frac{b_n}{b_{n+1}} - 1) > 0$. Prove that $\sum (-1)^n b_n$ converges.
Proof: Suppose $c = \lim_{n\to\infty} n\left(\frac{b_n}{b_{n+1}} - 1\right)$ with $c_n = n\left(\frac{b_n}{b_{n+1}} - 1\right)$, then we may assume $c_n > 0 \forall n \geqslant 1$, which implies $b_n > b_{n+1}$. If $\lim_{n\to\infty} b_n > 0$, then $\frac{b_n}{b_{n+1}} - 1 \to 0$ a contradiction. Hence $b_n \to 0$ so $\sum (-1)^n b_n$ converges.

11.3.7

Given $m \in \mathbb{Z}$, $\{a_n\} \to 0$ monotone, discuss whether $\sum a_n \sin \frac{n^2}{2m+1}\pi$ converges.
Solution: Note that for $b_n = \sin \frac{n^2}{2m+1}\pi$, $b_n = -b_{n+2m+1}$. Hence by Dirichlet's test $\sum a_n b_n$ converges.

11.3.8

Call $\sum a_n$ of bounded variation if $\sum |a_n - a_{n+1}| < \infty$. Prove that $\sum a_n$ is BV iff for any convergent $\sum b_n$, $\sum a_n b_n$ converges.
Proof: If $\sum a_n$ has bounded variation, then for any $\sum b_n < \infty$, $\sum a_n b_n = \sum (a_n - a_{n+1}) \sum_{i\leqslant n} b_i$. Clearly a_n BV implies a_n is bounded, so the boundary terms are bounded. Let $S_n = \sum_{i\leqslant n} b_i$, then $\sum |a_n b_n| \leqslant \sum |a_n - a_{n+1}| S_n$ where S_n convergent and $\sum |a_n - a_{n+1}|$ bounded. Hence $\sum a_n b_n$ converges.
Conversely, if $\sum |a_n - a_{n+1}| = \infty$, then define N_i inductively such that $\sum_{n=N_{i-1}+1}^{N_i} |a_n - a_{n+1}| \geqslant i$ and take the least such N_i. Expand $\sum_{n=N_{i-1}+1}^{N_i} |a_n - a_{n+1}|$ as $\sum_{n=N_{i-1}+1}^{N_i+1} c_i a_i$ with $c_n \in \{0, \pm 2\}$. Let $b_n = c_n/i^2$ for $N_{i-1} + 1 < i \leqslant N_i$, and $b_n = c_n/i^2 + c_n'/(i-1)^2$ for $i = N_{i-1} + 1$. Then $\sum b_n$ converges, while $\sum a_n b_n \geqslant \sum_i 1/i$ diverges.

11.3.9

Suppose $m \in \mathbb{N}$, prove that $\sum |\sin n^m|/n$ diverges.
Proof: Recall $\frac{1}{2\pi} n^m$ is equidistributed in $[0, 1)$. Consider $I = \left[\frac{1}{12}, \frac{5}{12}\right]$, and $A = \left\{n : \langle \frac{1}{2\pi} n^m \rangle \in I\right\}$, then $\#A \cap [0, n] \sim \frac{1}{3} n$ has positive density.
Note that $\sum |\sin n^m|/n \geqslant \frac{1}{2} \sum_{n\in A} \frac{1}{n}$. Suppose $A = \{n_1 < n_2 < \cdots\}$, then $\lim_{k\to\infty} n_k/k = 3$. Hence $\sum_{n\in A} \frac{1}{n} = \infty$ diverges.

11.3.10

Prove that $\sum \frac{\sin n^2}{n}$ converges.
Proof: This is a special case of Weyl's theorem. Consider Van de Corput's approach:

$$\begin{aligned}\left|\sum_{n=1}^{N} e^{in^2}\right|^2 &= \sum_{n,m} e^{in^2} e^{-im^2} = \sum_{|h|\leqslant N} \sum_{m,m+h\in[1,N]} e^{i(h^2+2hm)} \\ &= N + \sum_{h=1}^{N} e^{ih^2} \sum_{m=1}^{N-h} e^{2him} + \sum_{h=1}^{N} e^{ih^2} \sum_{m=h+1}^{N} e^{-2ihm}.\end{aligned}$$

Note that $\left|\sum_{m=1}^{M} e^{i\theta m}\right| \leqslant \min\{M, 1/\|\theta/2\pi\|\}$. Hence $|S(N)|^2 \leqslant \sum_{|h|<N} \min\{N, 1/\|h/\pi\|\}$.
Now consider the dyadic decomposition, let $A_n = \{|h| < N : \|h/\pi\| \in (2^{-n-1}, 2^{-n}]\}$ for $n = 0, 1, \cdots, K$ where $K = \lfloor \log_2 n \rfloor$, and $B = \{1, \cdots, N\} \setminus \bigcup A_k$. Then since h/π has finite irrationality measure, $|A_k| \leqslant 2^{-k}N + 1$ so

$$\sum_{|h|<N} \min\{N, 1/\|h/\pi\|\} \leqslant \sum_{k=0}^{K} 2^{k+1}|A_k| + N|B| \leqslant CN \log N.$$

Therefore $|S(N)| \leqslant C\sqrt{N \log N}$, hence by Abel summation the series $\sum \frac{\sin n^2}{n}$ converges.

11.4

11.4.1

IS there a series $\sum a_n$ such that $a_n \to 0$, $\sum a_n$ diverges, but it converges after adding brackets?
Solution: Yes: $a_{2n-1} = \frac{1}{n}, a_{2n} = -\frac{1}{n}$.

11.4.2

Prove that if $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum c_n$ diverges where $c_n = \sum a_m b_{n-m}$. ($a_n, b_n > 0$)
Proof: Trivial since $c_n \geqslant a_1 b_n$.

11.4.3

Construct $\sum a_n$ absolutely convergent and $\sum b_n$ conditionally convergent, such that $\sum d_n$ converges, where $d_n = a_1 b_n + a_2 b_n + \cdots + a_n b_n + a_n b_{n-1} + \cdots + a_n b_1$.
Solution: Let $a_n = (-1)^n 2^{-2^n}$, $b_n = (-1)^n/\sqrt{n}$. Then $d_n \geqslant \frac{1}{4\sqrt{n}} - \frac{1}{\sqrt{n}}(2^{-2^2} + \cdots) - n\sqrt{n} \cdot 2^{-2^n} \geqslant \frac{1}{8\sqrt{n}}$. Hence $\sum d_n$ converges.

11.5

11.5.2

11.5.4

Determine convergence:
(1) $\prod \frac{1}{n}$ diverges; (2) $\prod \frac{(n+1)^2}{n(n+2)}$ diverges.
(3) $\prod \sqrt{1 + \frac{1}{n}}$ diverges.

11.5.5

Determine convergence:
(1) $\prod \frac{n}{\sqrt{1+n^2}}$ converges.
(2) $\prod \left(\frac{n^2-1}{n^2+1}\right)^p$ converges.
(3) $\prod \sqrt{\log(n+x)-\log n}$ diverges.

11.5.6

Prove that if $\sum a_n^2 < \infty$ then $\prod \cos a_n$ converges.
Proof: $\cos a_n \in (-1,1)$, and $\sum (1-\cos a_n) = \frac{1}{2}\sum a_n^2 + O(a_n^4)$ converges, hence $\prod \cos a_n$ converges.

11.5.9

Prove that $\prod \frac{1}{e}\left(1+\frac{1}{n}\right)^n = 0$, and in particular, $\lim_{n\to\infty} \frac{n^n}{n!}e^{-n} = 0$.
Proof: Note that

$$\prod_{n=N}^{N+K} \frac{(n+1)^n}{n^n} = \frac{(N+K+1)^{N+K}}{N^N(N+1)\cdots(N+K)} = \frac{(N+K+1)^{N+K}N!}{(N+K)!N^N}.$$

By Stirling's formula, the product $\prod_{n=N}^{N+K} \frac{1}{e}\frac{(n+1)^n}{n^n}$ tends to 1 for large N.

11.5.10

Suppose $|a_n| < \frac{\pi}{4}$, $\sum a_n$ converges absolutely. Prove that $\prod \tan\left(\frac{\pi}{4}+a_n\right)$ converges.
Proof: Let $b_n = \tan a_n$, then $\tan\left(\frac{\pi}{4}+a_n\right) - 1 = \frac{1+b_n}{1-b_n} - 1 = \frac{2b_n}{1-b_n}$. For large n, $|a_n| < 1/4$ and $|b_n| \leqslant 2|a_n|$. Hence $\left|\frac{2b_n}{1-b_n}\right| \leqslant \frac{2|b_n|}{1-|b_n|} \leqslant \frac{4|a_n|}{1-2|a_n|} \leqslant 8|a_n|$. Therefore $\sum \tan\left(\frac{\pi}{4}+a_n\right) - 1$ converges, and $\prod \tan\left(\frac{\pi}{4}+a_n\right)$ converges.